



Multiphysics Modelling of Thermo-Concentration Coupled Acoustic Propagation in Salt Cavern Gas Storage Systems

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Abstract: Accurate prediction of acoustic propagation in salt cavern gas storage remains challenging due to the strong coupling between temperature and concentration fields in high-salinity environments. A multiphysics modelling framework is established by integrating piezoelectric, acoustic, and solid mechanics interactions with a temperature-dependent sound velocity formulation that accounts for concentration effects. The model is implemented in an axisymmetric configuration and evaluated under representative thermal and salinity conditions. The results demonstrate a pronounced nonlinear response of acoustic propagation to coupled temperature–concentration effects. Under elevated temperature and near-saturated brine conditions, a plateau-like behaviour in acoustic energy dissipation is observed, where further temperature increase leads to limited additional attenuation. This behaviour is governed by the competition between impedance matching efficiency and sound speed variation. Quantitative analysis indicates that reliance on room-temperature calibration may introduce systematic deviations of approximately 9% under high-temperature conditions. The study provides a physically grounded interpretation of acoustic behaviour under coupled-field conditions and offers a basis for improving the reliability of sonar-based detection in salt cavern environments. The proposed framework further contributes to the modelling of thermo-acoustic interactions in complex subsurface energy systems.

Keywords: Multiphysics coupling; Thermo–concentration interaction; Acoustic propagation; Salt cavern gas storage; Coupled-field modelling; Energy systems

1 Introduction

Energy security constitutes a fundamental component of national stability. As a strategic resource in the transition toward low-carbon energy systems, the reliability of natural gas supply directly influences economic continuity and infrastructure resilience [1]. In 2024, global natural gas consumption exceeded 4.2 trillion cubic meters, and ongoing geopolitical tensions have further underscored the necessity of developing large-scale and secure storage systems [2]. Among available technologies, salt cavern gas storage has been widely adopted due to its operational flexibility, high injection–withdrawal efficiency, and excellent sealing characteristics, with total working gas volumes surpassing 500 billion cubic meters worldwide [3].

Despite these advantages, long-term cyclic injection and withdrawal induce salt rock creep and structural instability, posing significant risks to cavern integrity. For instance, the Regina No. 5 salt cavern in Canada experienced two large-scale collapses resulting in approximately 45% volume loss [4], while notable deformation has also been reported in the Jintan storage facility in China after several years of operation [5]. Reliable monitoring of cavern morphology is therefore essential for ensuring safe and stable operation. Among available techniques, sonar-based cavity detection enables three-dimensional reconstruction through acoustic echo ranging and remains the only method capable of operating effectively in deep, high-salinity environments [6]. Commercial systems, such as the BSE sonar platform developed by SOCON, have demonstrated detection ranges of up to 300 meters and have been applied in engineering practice for decades [7].

The core component enabling these detection capabilities is the ultrasonic transducer, which converts electrical energy into acoustic waves and vice versa. As illustrated in Figure 1, ultrasonic transducers have found widespread

applications across diverse fields, ranging from ultrasound imaging for diagnostic purposes to industrial non-destructive testing for structural integrity assessment. In particular, phased array ultrasonic testing has enabled high-resolution imaging of complex geometries, while specialized sonar systems have been adapted for subsurface cavity detection in salt cavern gas storage facilities [8]. Despite these technological advances, the performance of ultrasonic transducers in extreme environments—characterized by high temperature, high pressure, and high salinity—remains insufficiently characterized.

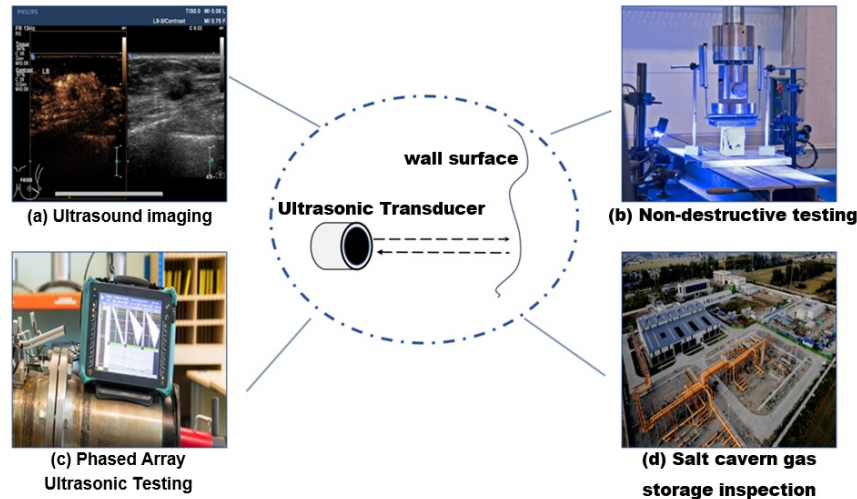


Figure 1. Applications of ultrasonic transducers across various fields

Existing studies have primarily focused on acoustic propagation in marine environments, whereas the underlying mechanisms in confined salt cavern systems remain insufficiently understood. In sonar-related applications, Wang et al. [9] developed a pressurized sonar testing device, Yang et al. [10] proposed a particle attenuation model based on scattering theory, and Yang et al. [11] established standardized ultrasonic detection procedures. Field applications, such as the Eminence salt dome gas storage facility, have demonstrated the capability of sonar monitoring to identify large-scale volume loss [12].

From the perspective of sound velocity modelling, classical formulations such as the Wilson and Chen–Millero equations are limited to standard seawater conditions (0–40 PSU), whereas salt cavern brine concentrations can reach up to 26 wt% (approximately 350 PSU), far exceeding their applicability range [13]. Kleis and Sanchez [14] proposed an empirical relationship for high-concentration NaCl solutions, later extended to high-pressure conditions by Kie lczyńska et al. [15]. Additional experimental studies have provided sound velocity data under elevated temperature and pressure [16, 17]. Nevertheless, these approaches do not explicitly account for the combined influence of temperature, pressure, and concentration, and the resulting coupling effects remain insufficiently characterised [18].

In terms of numerical modelling, multiphysics simulation tools such as COMSOL Multiphysics have been widely employed in acoustic–piezoelectric systems. Sheida et al. [19] identified and estimated the bandwidth characteristics of acoustic arrays composed of Tonpilz transducers in engineering sonar systems, providing insights into transducer array design for sonar applications. Fernandes et al. [20] validated time-explicit ultrasound propagation models, while Gandomi and Pearson [21], Silva and Santos [22] developed thermo-acoustic and quartz resonator models, respectively. Other studies have addressed acoustic attenuation, energy transmission, and high-intensity ultrasound systems through coupled modelling frameworks [23–27]. Parallel developments in salt cavern research have focused on thermo-mechanical behaviour, including creep, damage evolution, and stability analysis [28–33]. Although these studies provide valuable insights, they largely treat acoustic propagation and thermomechanical processes separately, without establishing a unified framework capable of capturing their coupled effects.

As a result, the behaviour of acoustic waves under high-temperature, near-saturated brine conditions—particularly at the scale relevant to field detection—remains inadequately understood. In particular, potential nonlinear phenomena arising from temperature–concentration coupling and their influence on acoustic attenuation and ranging accuracy have yet to be systematically clarified [34]. Previous studies on acoustic attenuation in suspensions and granular media have highlighted the sensitivity of wave propagation to thermophysical properties [35–40], but these findings have not been fully extended to salt cavern environments.

To address these limitations, a two-dimensional axisymmetric numerical model is developed to investigate acoustic propagation in salt cavern gas storage under coupled temperature–concentration conditions. The model integrates piezoelectric, acoustic, and solid mechanics interactions within a unified multiphysics framework, while

employing the Kleis–Sanchez formulation to describe sound velocity variations in high-concentration NaCl solutions. Parametric simulations are conducted over a temperature range of 20–60 °C to examine the combined influence of thermal and concentration fields on acoustic behaviour [41].

The analysis reveals a nonlinear modulation of acoustic propagation governed by the interaction between impedance matching efficiency and sound speed variation. In particular, a plateau-like behaviour in acoustic energy dissipation emerges under high-temperature and near-saturation conditions, indicating a transition in the dominant attenuation mechanisms. These findings provide a physically interpretable basis for improving sonar-based cavity detection and contribute to the broader understanding of coupled thermo-acoustic processes in complex subsurface energy systems.

2 Theoretical Model

2.1 Empirical Formula for Sound Velocity in High-Concentration NaCl Solutions

The accurate representation of sound velocity is essential for capturing acoustic propagation behaviour under coupled thermal and concentration conditions. Kleis and Sanchez conducted systematic measurements of sound velocity in high-concentration NaCl solutions over temperature and salinity ranges of 0–100 °C and 0–26 wt%, respectively, and proposed an empirical formulation suitable for concentrated brine environments [14]:

$$c(S, T) = c_w(T) + A(T) \cdot S + B(T) \cdot S^2 \quad (1)$$

where, $c(S, T)$ is the sound velocity in m/s, S is the NaCl mass fraction in wt%, T is the temperature in °C. $c_w(T)$ is the sound velocity in pure water, given by the following formula:

$$c_w(T) = 1402.39 + 5.04T - 0.058T^2 + 3.34wa \times 10^{-4}T^3 \quad (2)$$

$A(T)$ and $B(T)$ are salinity correction coefficients:

$$A(T) = 1.39 - 1.0 \times 10^{-3}T + 1.22 \times 10^{-4}T^2 \quad (3)$$

$$B(T) = -1.92 \times 10 - 4.5 \times 10^{-4}T \quad (4)$$

This formulation provides high accuracy within the concentration range of 0–26 wt% and temperature range of 0–100 °C, with relative errors below 0.5%. It is therefore well suited for describing sound velocity variations in near-saturated brine, where temperature–concentration coupling plays a dominant role in acoustic behaviour.

2.2 Numerical Simulation Model

To investigate acoustic propagation under coupled-field conditions, a two-dimensional axisymmetric numerical model is established using COMSOL Multiphysics 6.3 (as shown in Figure 2). The model represents a typical salt cavern gas storage system, including a brine domain with a 40 m characteristic diameter and a surrounding salt wall with a radius of 2 m.

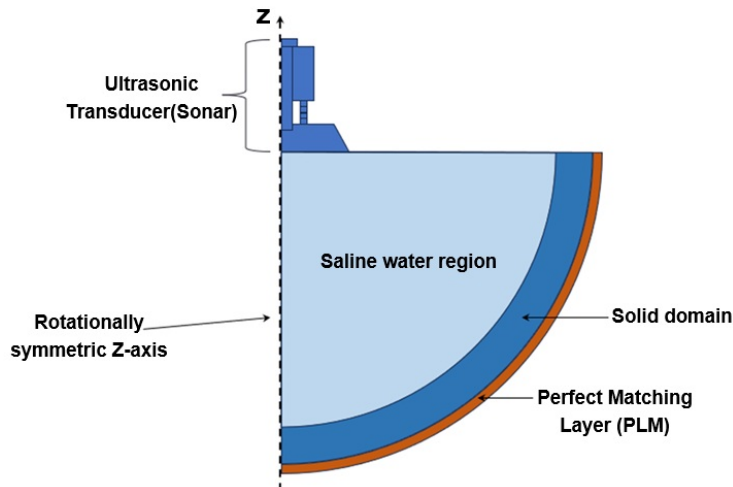


Figure 2. Axisymmetric section of the model's geometric structure

A multiphysics framework is adopted by integrating pressure acoustics, solid mechanics, and electrostatics. This configuration enables the simultaneous representation of piezoelectric excitation, acoustic wave propagation, and structural interaction, thereby capturing the coupled mechanisms governing energy transmission and dissipation in the system. Acoustic–structure interaction is implemented at the interfaces between the transducer, fluid domain, and solid boundary, allowing bidirectional coupling between acoustic pressure and structural response.

To ensure computational tractability, several simplifying assumptions are introduced: the brine density is assumed spatially uniform and symmetric, and the influence of auxiliary components such as probes and fixtures is neglected. The axisymmetric configuration allows efficient simulation of radial wave propagation while preserving the essential physical characteristics of the system.

2.3 Governing equations

The model requires solving hyperbolic partial differential equation groups for dual-field coupling in piezoelectric materials and coupled wave equations for dual domains in acoustic-structure boundaries. The piezoelectric constitutive equations describe the relationship between stress tensor S , strain tensor ε , electric field intensity E , and electric displacement vector D :

$$S = c : \varepsilon - E \cdot e \quad (5)$$

$$D = e : \varepsilon + \varepsilon_- \{0, vac\} (1 + \chi) E \quad (6)$$

where, c is the elastic stiffness matrix, e is the piezoelectric stress constant, $\varepsilon_- \{0, vac\}$ is the vacuum permittivity, and χ is the electric susceptibility. The acoustic-structure coupling boundary conditions are completed through solid mechanics and pressure acoustics modules. Displacement is numerically differentiated to obtain boundary acceleration, and then the total sound pressure p_t is calculated through the pressure acoustics governing equation:

$$1/(\rho c^2) \cdot \partial^2 p_t / \partial t^2 + \nabla \cdot (-1/\rho (\nabla p_t - q_d)) = Q_m \quad (7)$$

where, c denotes the speed of sound, p_t the total acoustic pressure, t time, q_d the dipole source term, and Q_m the mass source term. The acoustic pressure at the interface is mapped to structural forces within the solid domain, while enforcing continuity of normal acceleration across the fluid–solid boundary to achieve closed-loop acoustic–structural coupling.

2.4 Thermophysical parameters and initial conditions

Thermophysical parameters are defined based on material properties relevant to salt cavern systems. The brine properties incorporate temperature-dependent sound velocity through the empirical formulation described above, while the mechanical properties of aluminum, steel, and salt rock are obtained from the COMSOL material library (as shown in Table 1).

Table 1. Thermophysical properties of materials

Parameter Name	Material Parameters	Expression
brine	K	$T = t + 273.15$
	m/s	$c(S, T)$
	Density (kg/m ³)	2700
Aluminum	Young's modulus (GPa)	70
	Poisson's ratio	0.33
	Density (kg/m ³)	7850
Steel	Young's modulus (GPa)	205
	Poisson's ratio	0.28
	Density (kg/m ³)	2300
Salt wall	Young's modulus (GPa)	15
	Poisson's ratio	0.28

The initial conditions are specified to capture representative operational ranges of temperature and concentration in salt cavern environments (as shown in Table 2). These parameter ranges enable systematic evaluation of temperature–concentration coupling effects on acoustic propagation behaviour.

Table 2. Initial condition set value

Parameter Name	Unit	Value
Temperature matrix	°C	[20,30,40,50,60]
Concentration matrix	wt%	[6.5,13,19.5,26]

2.5 Mesh Division and Solver Settings

The computational domain is discretized into three regions: the transducer, the brine domain, and the surrounding solid domain. A refined triangular mesh is employed in the transducer region to accurately resolve piezoelectric responses, while structured quadrilateral meshes are applied in the fluid and solid domains to ensure a balance between numerical accuracy and computational efficiency.

Mesh resolution is selected based on acoustic wavelength considerations to adequately capture wave propagation characteristics across different media. The characteristic mesh sizes adopted in each region are summarized in Table 3. These settings ensure sufficient spatial resolution for representing coupled acoustic–structural interactions while maintaining computational stability.

The solver configuration is designed to achieve stable convergence for the coupled multiphysics system, enabling reliable simulation of acoustic behaviour under varying temperature and concentration conditions.

Table 3. Grid cell size

Name	Maximum Cell	Unit
Ultrasonic Transducer	5	mm
Saltwater area	$c / 10[\text{kHz}] / 6$	mm
Solid domain	$c_R / 10[\text{kHz}] / 6$	mm

3 Simulation Results and Analysis

3.1 Temperature-Concentration Coupled Field Simulation Results

Parametric simulations were conducted to investigate acoustic propagation under coupled temperature-concentration conditions within a representative thermal range of 20–60 °C. The transient acoustic pressure responses at the observation point were evaluated across different NaCl mass fractions, and the corresponding waveform variations at representative temperatures are presented in Figures 3–7.

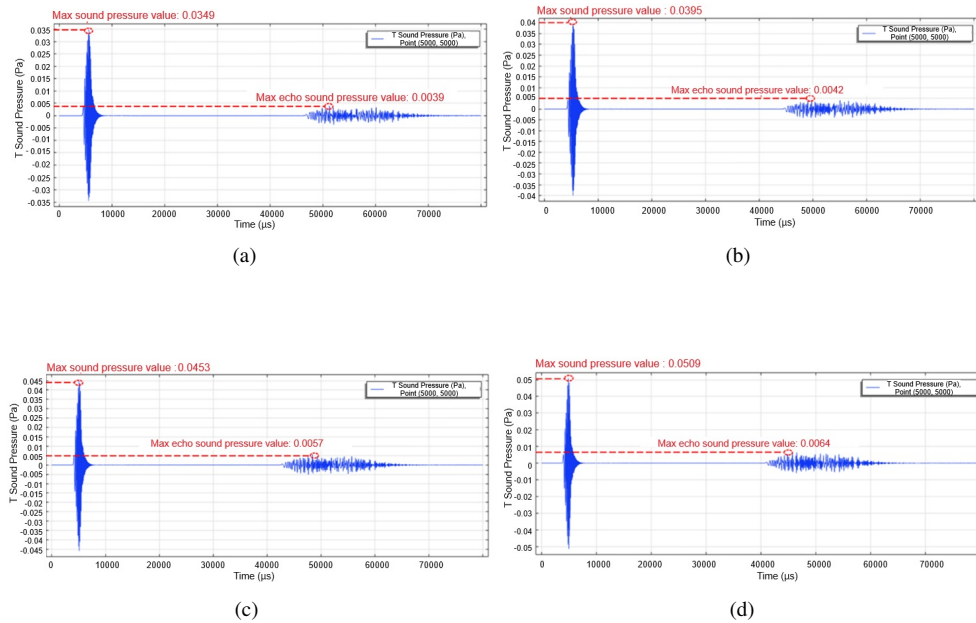


Figure 3. The effect of concentration gradient on waveform at 20 °C: (a) 6.5 wt%; (b) 13 wt%; (c) 19.5 wt%; (d) 26 wt%

At each fixed temperature condition, the acoustic waveform exhibits a clear dependence on concentration. As the NaCl mass fraction increases, both the amplitude of the transient sound pressure and the corresponding echo response show a consistent increase, indicating enhanced acoustic energy transmission in higher-concentration media. This trend is observed across all temperature conditions, although the rate of increase varies with temperature.

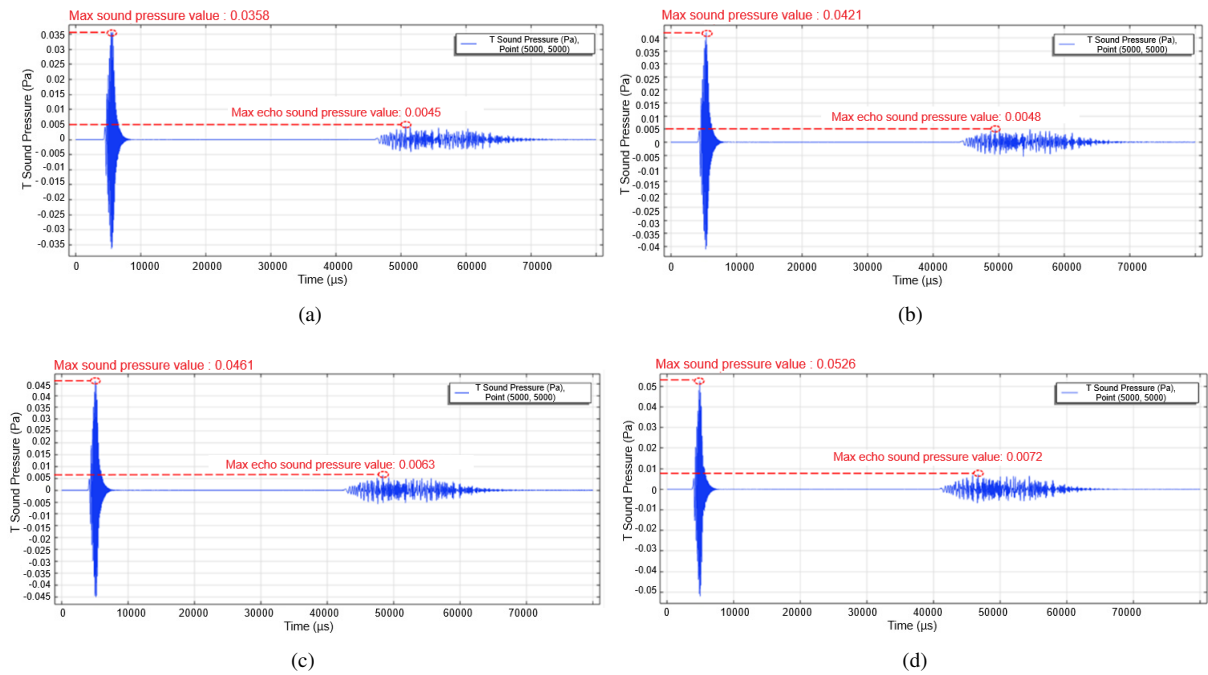


Figure 4. The effect of concentration gradient on waveform at 30 °C: (a) 6.5 wt%; (b) 13 wt%; (c) 19.5 wt%; (d) 26 wt%

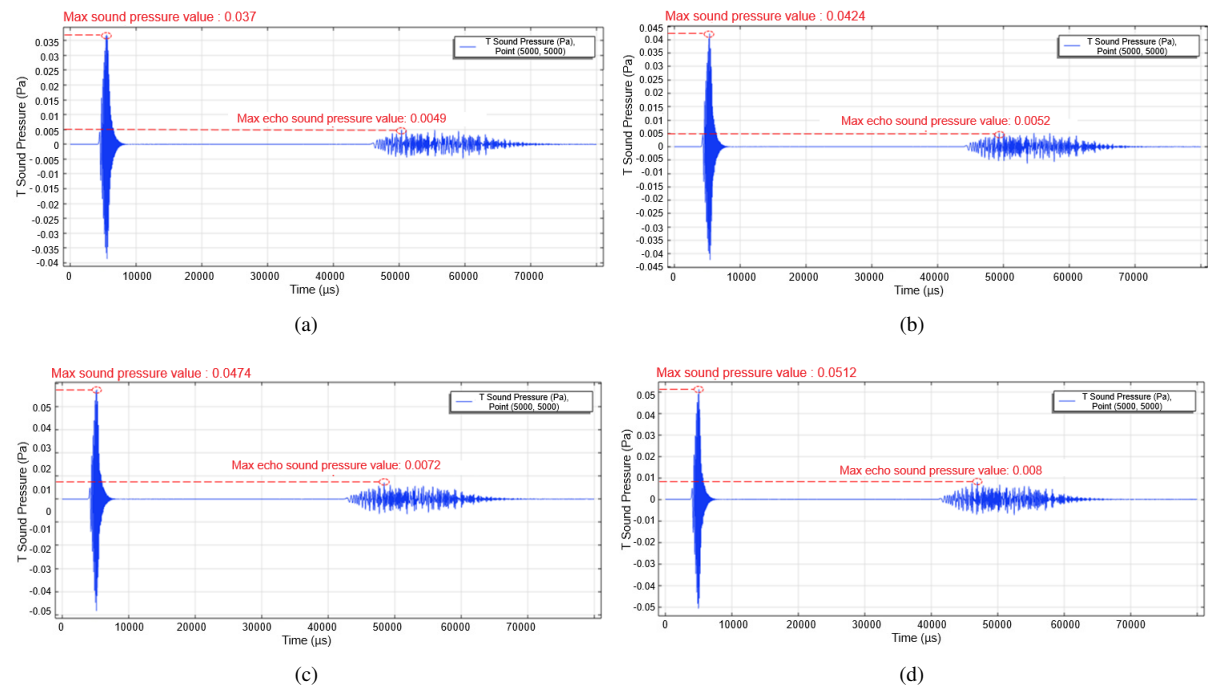


Figure 5. The effect of concentration gradient on waveform at 40 °C: (a) 6.5 wt%; (b) 13 wt%; (c) 19.5 wt%; (d) 26 wt%

When comparing different temperature levels, a systematic variation in acoustic response becomes evident. At lower temperatures (20–30 °C), the waveform amplitudes remain relatively moderate, while an increase in temperature generally leads to higher acoustic pressure levels. However, this temperature-induced enhancement is not uniform across all concentration ranges. In particular, at higher concentrations approaching near-saturation conditions, the differences between temperature cases become less pronounced, suggesting a transition in the governing mechanisms of acoustic propagation.

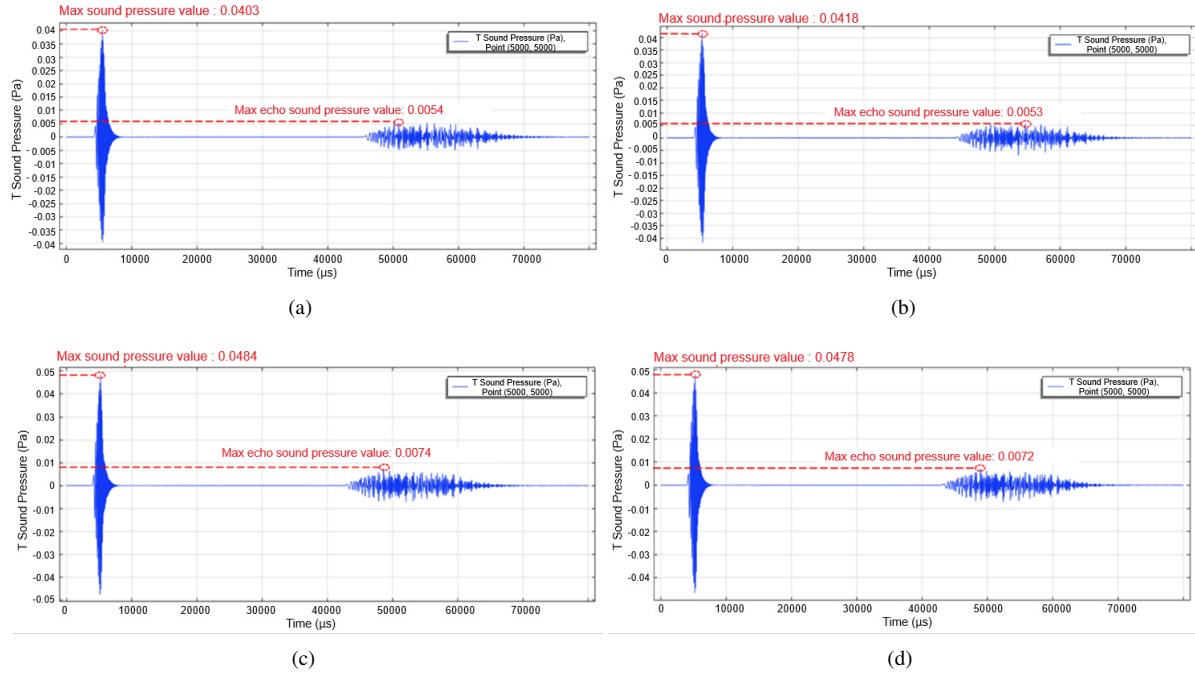


Figure 6. The effect of concentration gradient on waveform at 50 °C: (a) 6.5 wt%; (b) 13 wt%; (c) 19.5 wt%; (d) 26 wt%

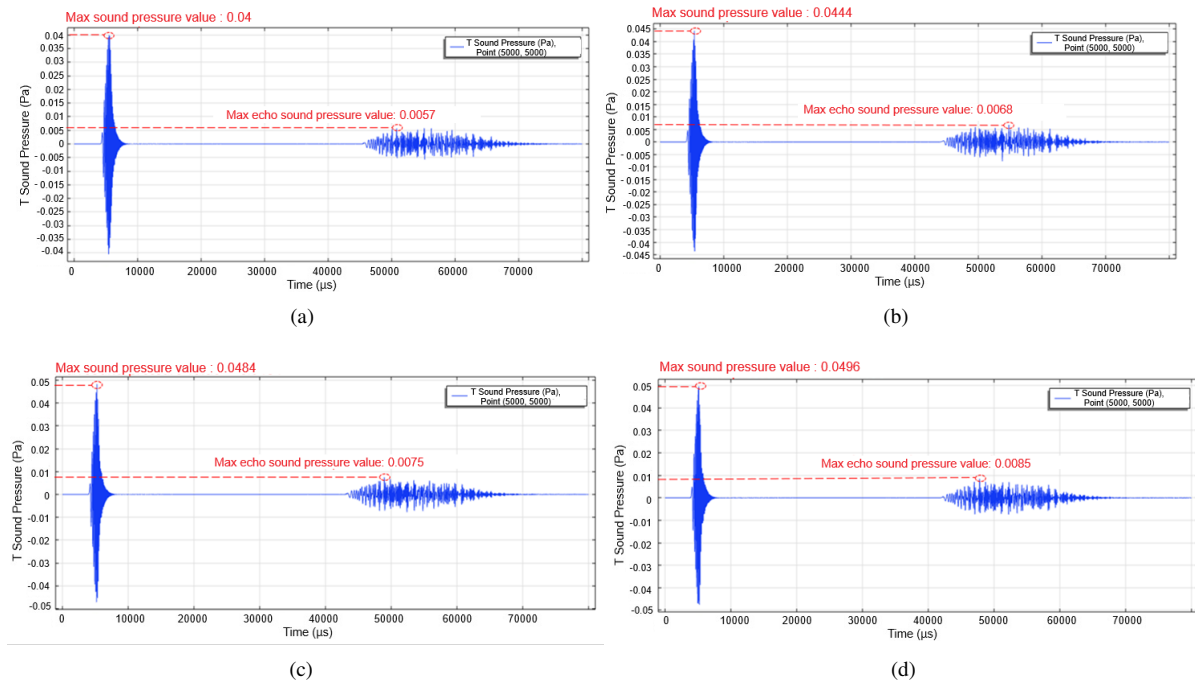


Figure 7. The effect of concentration gradient on waveform at 60 °C: (a) 6.5 wt%; (b) 13 wt%; (c) 19.5 wt%; (d) 26 wt%

The maximum acoustic pressure values and corresponding dissipation characteristics are summarised in Table 4 and Table 5. These quantitative results provide the basis for further analysis of the coupled influence of temperature and concentration on acoustic behaviour.

Table 4. Maximum sound pressure values under temperature gradient influence

Mass Fraction (wt%)	20 °C	30 °C	40 °C	50 °C	60 °C
6.5	0.0349	0.0358	0.037	0.0403	0.04
13	0.0395	0.0421	0.0424	0.0418	0.0444
19.5	0.0453	0.0461	0.0474	0.0484	0.0484
26	0.0509	0.0526	0.0512	0.0478	0.0496

Table 5. Sound pressure dissipation value under temperature gradient influence

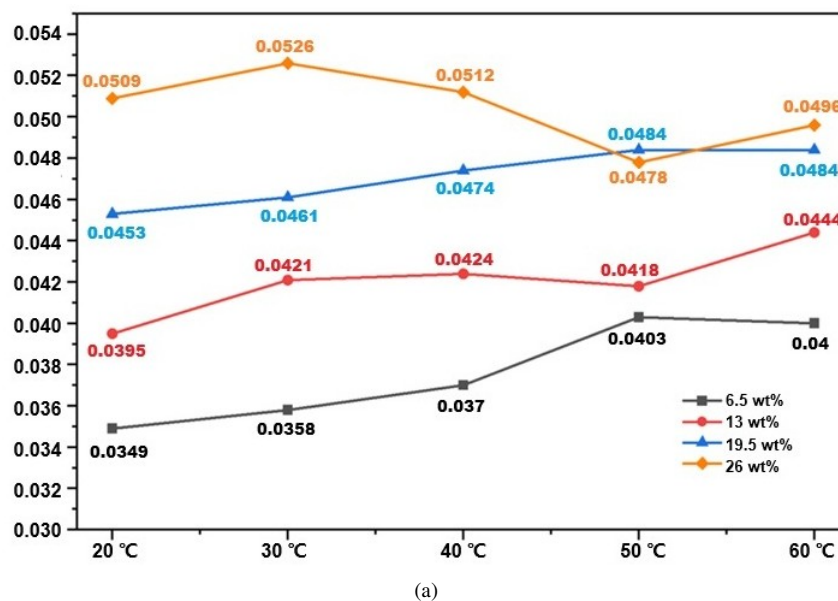
Mass Fraction (wt%)	20 °C	30 °C	40 °C	50 °C	60 °C
6.5	0.031	0.0313	0.0321	0.0349	0.0343
13	0.0353	0.0373	0.0372	0.0365	0.0376
19.5	0.0396	0.0398	0.0402	0.041	0.0409
26	0.0445	0.0454	0.0432	0.0406	0.0411

3.2 Results Analysis

The variation of acoustic pressure and dissipation with concentration and temperature is illustrated in Figure 8, based on the quantitative results summarised in Tables 4 and 5. The coupled influence of temperature and concentration gives rise to a distinctly nonlinear response in acoustic propagation behaviour.

In the low to moderate concentration range (6.5–19.5 wt%), the acoustic pressure amplitude increases approximately monotonically with temperature. Under these conditions, higher temperatures generally correspond to stronger acoustic responses, reflecting the combined effects of increased sound velocity and improved impedance matching between the transducer and the surrounding medium. The relative ordering of the temperature curves remains stable, indicating a predictable thermo-acoustic response regime.

As the concentration approaches the near-saturation level (26 wt%), a transition in behaviour becomes evident. The curves corresponding to different temperatures exhibit convergence and local crossover, forming a characteristic “scissor-like” pattern. For instance, at 50 °C, the acoustic pressure decreases slightly from 0.0484 Pa to 0.0478 Pa, whereas at 30 °C it continues to increase, reaching 0.0526 Pa. This shift indicates a change in the dominant mechanisms governing acoustic propagation under coupled-field conditions.



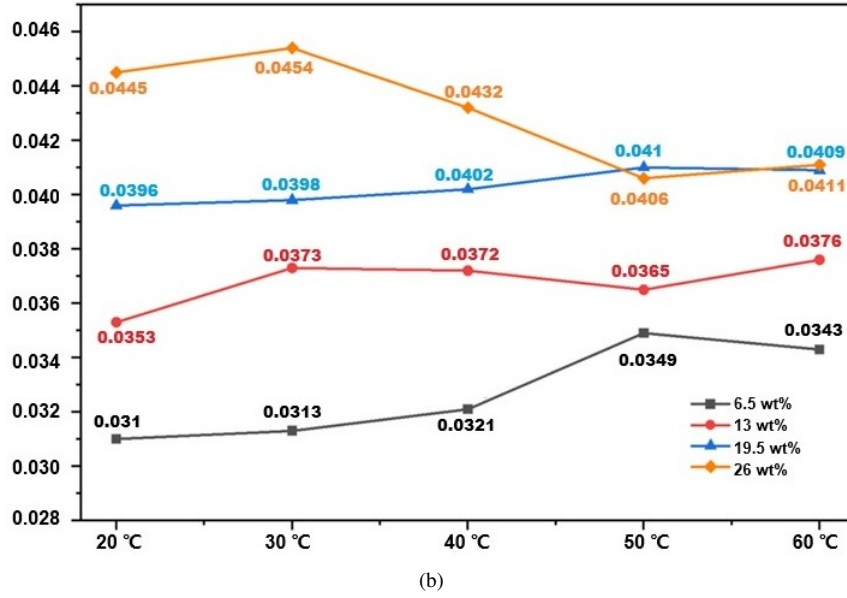


Figure 8. Variation of sound pressure and dissipation with concentration and temperature: (a) MSPV trend with concentration under temperature gradient; (b) MSPV trend with concentration under temperature gradient

A comparable transition is observed in the dissipation behaviour. Within the 20–40 °C range, the dissipation values increase steadily with concentration, and the corresponding curves remain nearly parallel. In contrast, at higher temperatures (50–60 °C), the dissipation curves exhibit a pronounced plateau in the near-saturation region. At 26 wt%, the dissipation values for 50 °C and 60 °C are approximately 0.0406 Pa and 0.0411 Pa, respectively, which are about 6% lower than the value at 40 °C. This plateau indicates that further increases in temperature no longer lead to proportional increases in energy dissipation.

This behaviour can be attributed to the competing interaction between multiple physical mechanisms. On the one hand, increasing temperature enhances impedance matching and promotes acoustic energy injection. On the other hand, temperature-induced reductions in fluid viscosity decrease viscous losses, while the associated decrease in sound velocity contributes to attenuation. The dynamic balance between these competing effects leads to saturation in dissipation under high-temperature, high-concentration conditions.

From an engineering perspective, this nonlinear behaviour has direct implications for sonar-based cavity detection. If calibration based on room-temperature conditions is applied to high-temperature environments, a systematic overestimation of approximately 9.1% may occur under near-saturated conditions. This highlights the necessity of adopting segmented temperature compensation strategies. In shallow regions characterised by lower temperature and concentration, transmission power can be reduced to improve efficiency, whereas in deeper regions with elevated temperature and near-saturated brine, dedicated correction schemes are required to account for the coexistence of natural attenuation and dissipation saturation.

4 Conclusions

Acoustic propagation in salt cavern gas storage is strongly influenced by the coupled effects of temperature and concentration, particularly under high-temperature and near-saturated brine conditions. The present study establishes a multiphysics modelling framework integrating piezoelectric, acoustic, and solid mechanics interactions to investigate these coupled processes within a unified computational setting.

The results demonstrate that acoustic propagation exhibits a pronounced nonlinear response to temperature-concentration coupling. In low to moderate concentration regimes, acoustic pressure increases monotonically with temperature, reflecting enhanced impedance matching and energy transmission. However, as the concentration approaches near-saturation conditions, a transition in behaviour occurs. A characteristic “scissor-like” pattern emerges in the acoustic pressure response, indicating a shift in the dominant mechanisms governing wave propagation.

A key finding of this study is the identification of a plateau-like behaviour in acoustic energy dissipation under high-temperature and high-concentration conditions. When the temperature exceeds approximately 40 °C and the concentration approaches saturation, further increases in temperature result in only limited additional dissipation. This behaviour arises from the competing interaction between impedance matching effects and sound velocity reduction, which leads to a dynamic balance in energy transfer processes.

Quantitative analysis indicates that the use of room-temperature calibration under such conditions may introduce systematic deviations of approximately 9%, highlighting the limitations of conventional calibration approaches. This finding underscores the necessity of adopting segmented temperature compensation strategies in practical sonar-based detection systems.

From a modelling perspective, the proposed multiphysics framework provides a consistent approach for capturing coupled thermo-acoustic behaviour in complex subsurface environments. The results offer a physically interpretable basis for understanding acoustic propagation under extreme conditions and provide guidance for the optimisation of transducer parameters and detection strategies in salt cavern applications.

Future work may extend the present model to three-dimensional geometries and incorporate additional coupled effects, such as pressure-dependent behaviour, to further improve predictive capability and support the development of advanced compensation algorithms for high-precision sonar systems.

Author Contributions

Conceptualization, Yu Wang; methodology, Yang Wang and Yu Wang; writing—original draft preparation, Yang Wang; writing—review and editing, Yu Wang; visualization, Yang Wang; funding acquisition, Yu Wang. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data that supports the findings of this study are available on request from the corresponding author. The data is not publicly available due to privacy or ethical restrictions.

Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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